Deliverable 1: Requirements Analysis

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I. Project Overview

The project 'Home Food Marketplace' is the development of a simple web application designed as a community-driven marketplace for homemade food, similar to DoorDash. Its primary goal is to connect individual merchants looking to monetize their cooking skills with local buyers seeking affordable, flexible, home-cooked meals. Key features include a search function based on general food category and a product feed that supports images, and like/save interactions. The unique selling proposition is to strengthen the local economy and foster a sense of community through food sharing.

II. Requirements Explanation

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The application's requirements are fundamentally organized around four distinct user roles, each with specific functional responsibilities and access privileges. All user roles must support basic authentication features, including registration, and login/logout. The delivery driver and admin roles proposed in the project proposal were removed to reduce the scope of the project.

The system's core functional requirements are defined by these roles:

- **Primary User Role (Customer):** Requirements center on the purchasing journey. Customers must be able to search for and view stores and products, place orders, complete payments, and profile management (view/edit).
- **Secondary User Roles (Merchant):** Requires capabilities to manage their inventory (create/manage menu items) and handle orders (view, accept, and cancel), along with editing restaurant information.

III. Functional Requirements/Use Cases

This section documents all functional requirements as a collection of use cases, which covers all mandatory user roles.

II.1 General User Use Cases

Use Case ID	UC-G01

Use Case Name	User Login
Actor	User (Registered Customer, Merchant, or Delivery Driver)
Description	This use case allows a user to authenticate and gain authorized access to the application based on their assigned role.
Pre-conditions	User is on the Index Page. System is operational.
Post-conditions (Success)	User is authenticated. Login attempts counter is reset. System redirects user to their respective dashboard.
Post-conditions (Failure)	User remains on the Login Page. Error message displayed. Login attempts counter is incremented.
Main Flow (Basic Flow)	1. User clicks "Login" button on the Index Page. 2. System displays the Login Page with fields for Email and Password. 3. User enters a registered Email Address and Password. 4. User clicks "Sign In" button. 5. System validates the provided credentials against the user database. 6. System grants access and directs the user to their appropriate dashboard (e.g., Customer Dashboard, Merchant Dashboard).
Alternative Flows	**Alternative Flow 1: Forgot Password.** Trigger: User clicks the "Forgot Password" link. 3a. System directs the user to the Password Reset page. 3b. User enters their email address and submits. 3c. System sends a password reset link to the email address.

Exception Flow 1	**Exception Flow 1: Invalid Credentials.** Trigger: System validation fails (Email/Password combination does not match records). 5a. System displays error message: "Invalid email or password. Please try again." 5b. Flow returns to step 3.
Business Rules	Password input is case-sensitive.
Special Requirements	Authentication must use secure hashing for password storage.

Use Case ID	**UC-G02**
Use Case Name	User Registration (Sign Up)
Actor	Guest User
Description	This use case allows a new user to create an account and specify their primary role (Customer, Merchant, or Delivery Driver).
Pre-conditions	User is on the Index Page. User does not have an existing account.
Post-conditions (Success)	A new User account is created. User is automatically logged in. User is redirected to their respective dashboard or a profile setup screen.
Post-conditions (Failure)	Account creation is prevented. User remains on the Registration Page. Error message displayed.

Main Flow (Basic Flow)	1. User clicks "Sign Up Free" button on the Index Page. 2. System displays the Registration Page. 3. User enters required information: 3.1 First Name 3.2 Last Name 3.3 Email Address 3.4 Password 3.5 Confirmation Password. 4. User selects their intended user role (Customer, Merchant, or Delivery Driver). 5. User checks the "Agree to Terms & Conditions" box. 6. User clicks the "Register" button. 7. System performs input validation (e.g., matching passwords, valid email format). 8. System checks the database to ensure the email address is unique. 9. System creates the new user record, automatically logs the user in, and redirects them.
Alternative Flows 1: Access through Login Page	Trigger: User is on the login page and wants to register for an account. 2a. User clicks on the ‘Sign Up Free’ button in the login page. 2b. User is redirected to the register page. 2c. Flow continues to step 3.
Exception Flows	**Exception Flow 1: Email Already Exists.** Trigger: System finds a duplicate email address in the database during validation. 8a. System displays error message: "This email address is already registered. Please log in or use a different email." 8b. Flow returns to step 3.

Exception Flow 2: Password Mismatch.	**Trigger**: Password and Confirmation Password fields do not match. 7a. System highlights password fields and displays error message: "Passwords do not match." 7b. Flow returns to step 3.
Business Rules	Password must meet complexity requirements (e.g., minimum length, uppercase letter, number). Email address must be unique across the platform. The user must select one primary role.
Special Requirements	User must accept Terms & Conditions before registering.

II.2 Primary User Role Use Cases: [Customer]

Use Case ID	**UC-C01**
Use Case Name	Browse and Search Food Listings
Actor	Customer
Description	This use case describes how a customer discovers cooks or dishes by browsing listings, searching with keywords, or selecting cuisine categories.
Pre-conditions	* Customer has access to the Browse page. * System displays available food listings.
Post-conditions (Success)	* Customer views food listings that match their interest. * Customer navigates to a restaurant detail page.

Post-conditions (Failure)	* Customer remains on the Browse page with a message indicating no results.
Main Flow (Basic Flow)	1. Customer navigates to the Browse page. 2. System displays featured cooks and food listings. 3. Customer enters a keyword into the search bar. 4. Customer clicks Search. 5. System displays search results that match the query. 6. Customer selects a listing. 7. System displays the restaurant detail page for the selected listing alongside the following information: 7a. List of all products (with name, price, and description) 7b. Restaurant address (Merchant address)
Alternative Flow 1: Browse by Category	Trigger: Customer prefers browsing categories. 3a. Customer selects a cuisine category (e.g., Italian, Korean, Vegan). 3b. System filters the listings according to the selected category. 3c. Customer selects a listing. 3d. Flow continues to step 7 of the Main Flow.
Alternative Flow 2: Browse Without Searching	Trigger: Customer is casually exploring available food listings. 3a. Customer scrolls through the listing feed. 3b. System loads additional listings dynamically. 3c. Customer selects a listing. 3d. Flow continues to step 7 of the Main Flow.
Exception Flow 1: No Search Results	Trigger: The search query returns zero results. 5a. System checks for matching listings. 5b. System detects no matching results. 5c. System displays a “No results found” message. 5d. Customer revises the search query.

	5e. Flow returns to step 3.
Business Rules	* Search should support partial keyword matching. * Listings should display rating, cuisine type, and estimated delivery time.
Frequency of Use	High - This is the entry point for almost every user session where a purchase is intended.
Priority	Critical - Core to the application's purpose of connecting users with food.

Use Case ID	**UC-C02**
Use Case Name	Manage Cart
Actor	Customer
Description	This use case describes how a customer reviews items in the cart, updates quantities, or removes items before checkout.
Pre-conditions	* Customer has selected at least one food item. * Cart page is accessible.
Post-conditions (Success)	* Cart reflects updated items or quantities. * Order total is recalculated.
Post-conditions (Failure)	* Cart remains unchanged.
Main Flow (Basic Flow)	1. Customer navigates to the Cart page. 2. System displays all items currently in the cart. 3. Customer adjusts item quantities. 3.1 Customer clicks + to increase quantity. 3.2 Customer clicks − to decrease quantity. 3.3 System updates the quantity displayed. 4. System recalculates the total price. 5. Customer reviews the updated cart

	summary.
Alternative Flow 1: Remove Item	Trigger: Customer decides to remove an item. 3a. Customer clicks the remove icon beside an item. 3b. System removes the item from the cart. 3c. System recalculates the total price. 3d. Flow continues to step 5.
Exception Flow 1: Cart Becomes Empty	Trigger: All items are removed. 3a. System detects that the cart is empty. 3b. System displays a message indicating the cart is empty. 3c. Customer may return to the browsing page.
Business Rules	* Quantity must be at least 1. * Cart totals update automatically when quantities change.
Frequency of Use	High - Used for every successful transaction before proceeding to checkout.
Priority	Critical - Directly precedes order placement and revenue generation.

Use Case ID	**UC-C03**
Use Case Name	Checkout and Place Order
Actor	Customer
Description	This use case describes how a customer completes the order by providing delivery information and confirming payment.
Pre-conditions	* Customer is logged in. * Cart contains at least one item.
Post-conditions (Success)	* Order is created with a unique order ID. * Order confirmation page is displayed.

Post-conditions (Failure)	* Customer remains on the checkout page with error messages.
Main Flow (Basic Flow)	1. Customer clicks Proceed to Checkout from the Cart page. 2. System displays the Checkout page. 3. Customer enters delivery information. 3.1 First name 3.2 Last name 3.3 Street address 3.4 Apartment or suite (optional) 3.5 City 3.6 State 3.7 ZIP code 3.8 Phone number 3.9 Delivery notes (optional) 4. Customer selects fulfillment option. 5. Customer reviews order summary. 6. Customer clicks Place Order. 7. System processes the order and creates an order record. 8. System displays the order confirmation page.
Alternative Flow 1: Pickup Instead of Delivery	Trigger: Customer prefers to pick up the order. 4a. Customer selects Pickup option. 4b. System updates order summary. 4c. Flow continues to step 5.
Exception Flow 1: Invalid Address	Trigger: Delivery address information is incomplete. 3a. System detects missing or invalid fields. 3b. System highlights the problematic fields. 3c. Customer corrects the address information. 3d. Flow returns to step 3.
Exception Flow 2: Payment Failure	Trigger: Payment processing fails. 7a. System detects payment error. 7b. System displays a payment failure message. 7c. Customer attempts the payment again.

	7d. Flow returns to step 5.
Business Rules	* Total order amount must be greater than zero. * Required fields must be validated before order placement.
Frequency of Use	High - Used for every successful customer transaction.
Priority	Critical - This is the central function for generating sales and revenue for the application.

Use Case ID	**UC-C04**
Use Case Name	View Order History
Actor	Customer
Description	This use case describes how a customer reviews previously completed orders.
Pre-conditions	* Customer has previous orders.
Post-conditions	* Customer can view details of previous orders.
Main Flow (Basic Flow)	1. Customer navigates to the Dashboard. 2. Customer selects Order History. 3. System displays a list of previous orders. 4. Customer clicks on an order from the list. 5. System displays detailed information about the selected order.
Alternative Flow 1: Reorder Items	**Trigger**: Customer wants to reorder previously purchased items. 4a. Customer clicks Reorder. 4b. System adds the same items to the cart. 4c. Customer proceeds to checkout.
Exception Flow	None

Business Rules	* Orders should be sortable by date or time range. * Order totals and item details must be displayed.
Frequency of Use	Moderate - Used occasionally for expense tracking or initiating reorders.
Priority	Medium - Important for account usability and encouraging repeat business, but not critical to basic transaction flow.

Use Case ID	**UC-C05**
Use Case Name	Edit User Infomation
Actor	Customer
Description	Allows a user to update their profile information and/or primary street address used for order delivery.
Pre-conditions	User is logged in. User is on the Profile Management screen.
Post-conditions (Success)	User's profile information and/or delivery address is updated. Success notification is displayed.
Post-conditions (Failure)	Records remains unchanged. Error message displayed.
Main Flow (Basic Flow)	1. User views the Profile Management screen. 2. User clicks the 'Edit' button. 3. System renders fields as editable including: 3.1 First Name 3.2 Last Name 3.3 Phone Number 3.4 Date of Birth 3.5 Street 3.6 City

	3.7 State 3.8 Zip 4. User modifies one or more of the fields. 5. User clicks the 'Save' button. 6. System validates that all mandatory address fields are populated. 7. System updates the user's record. 8. System displays a confirmation message and returns to the read-only profile view.
Alternative Flows	**Alternative Flow 1: Cancel Edit.** Trigger: User decides not to save changes. 4a. User clicks the 'Cancel' button. 4b. System discards any changes and returns to the read-only profile view. 4c. Flow aborts.
Exception Flows	**Exception Flow 1: Missing Mandatory Field.** Trigger: System validation detects a required field (e.g., City) is blank. 6a. System displays error message: "All fields are required." 6b. System highlights the missing field. 6c. Flow returns to step 4.
Business Rules	All fields are mandatory for a complete profile.
Special Requirements	Input should support auto-completion or verification against a recognized geographic service.

II.3 Secondary User Role Use Cases: [Merchant]

Use Case ID	UC-M01
Use Case Name	Add Menu Item
Actor	Registered Merchant
Description	Allow Merchants to add new Items to their menu

Pre-conditions	Logged into a merchant account Merchant must have valid email address
Post-conditions(success)	New Menu Item added to Merchants menu
Post-conditions(Failure)	Item not added to merchants menu Error Message is displayed
Main Flow(Basic Flow)	1. Merchant Presses add item button 2. A form displays 3. Merchant inputs info in the form 3.1. ingredients 3.2. calories 3.3. servings 3.4. name 3.5. price 4. Merchant Presses “Complete Item” Button 5. System validates input 6. System creates new item 7. System updates menu
Alternative Flow 1: Save as Draft	**Trigger:** User Presses “Save as Draft” instead of “Complete Item” 4a. Merchant Presses “Save as Draft” 4b. System Validates input for correct inputs and skips missing inputs 4c. System saves item as a draft 4d. merchants draft list is updated by the system

Exception Flow 1: Invalid Price	**Trigger:** Price not > $0 or is not in numerical format 3a. Error Message Is Displayed 3b. price input box highlighted 3c. Merchant corrects price 3d. Flow returns to step 4 of Main Flow
Exception Flow 2: Invalid Item Name	**Trigger:** Item name already appears in merchant's menu 3a. Error Message Is Displayed 3b. Name Input Box highlighted 3c. merchant corrects name 3d. flow returns to step 4 of Main flow
Exception Flow 3: System Error prevents adding item	**Trigger:** System fails at some point in process not fully completing 4a. Error Message displays "System Error Item Not Created" 4b. Any previous parts of the process are undone (All or nothing function)
Exception Flow 4: Invalid Calorie Input	**Trigger:** Item calorie < 0 3a. Error Message Is Displayed 3b. calorie input box highlighted 3c. Merchant corrects calories 3d. Flow returns to step 4 of Main Flow

Business Rules	· Item name must be unique within a merchants menu · Price must be positive · Calories must be positive · Draft Items won't appear on live menu · All fields must be validated before being saved as a menu item
Frequency of Use	High-This is a critical function without which the rest of the app will not work
Priority	Critical – Core Functionality

Use Case ID	UC-M02
Use Case Name	Delete Menu Item
Actor	Registered Merchant
Description	Allows Merchant to delete an item currently on their menu
Pre-conditions	Merchant Is Email Verified Item Exists on Menu
Post-conditions(success)	System Updated Item No Longer Exists
Post-conditions(Failure)	System Item Still Exists Error Message Displayed

Main Flow(Basic Flow)	1. User Presses “Delete Item” Button on Menu Page 2. Dialog box displays asking for confirmation of Deletion 3. User Presses “Delete” Button 4. Dialog Box Closes 5. Item is Deleted from the Menu 6. Menu Page automatically Refreshes
Alternative Flow 1: User Cancels Deletion	**Trigger:** User Presses “Cancel” Button on Dialog box instead of “Delete” 3a. User Presses “Cancel” 3b. Dialog Box Closes 3c. Menu Items remain Unchanged
Exception Flow 1: Item deleted before process completed	**Trigger**: Item Deleted by other process before item deletion completed 4a. Error message displays “Item no longer exists” 4b. Dialog box Closes 4c. Menu Page automatically Refreshes
Exception Flow 2: System Error prevents deletion	**Trigger:** System fails at some point in process not fully deleting 4a. Error Message displays “System Error Item Not Deleted” 4b. Any previous parts of the process are undone (All or nothing function) 4c. Dialog Box Closes

Business Rules	· Item must be on Menu before deletion · If last item on menu deleted system disables “Open Store” function · Deletion must be confirmed by merchant before system removes the item
Frequency of Use	High – Used whenever there is a change on Menu
Priority	High- Core Functions of Store still functional but function is needed for improved merchant experience

Use Case ID	UC-M03
Use Case Name	Edit Item
Actor	Registered Merchant
Description	Allows a merchant to edit items on their menu or in their item drafts
Pre-conditions	Merchant Is Email Verified Item Exists on Menu/Draft List
Post-conditions(success)	Item Info is updated in System
Post-conditions(Failure)	Item Remains unchanged Error Message Displayed

Main Flow(Basic Flow)	1. User Presses “Edit Item Button” 2. Item Form appears populated with current info 3. User inputs new updated info 3.1. ingredients 3.2. calories 3.3. servings 3.4. name 3.5. price 4. user presses “Complete Item” Button 5. Item Info updated on Merchant’s Menu in System 6. Menu Page automatically refreshes
Alternative Flow 1: Save as Draft(already published menu item)	**Trigger:** User Presses “Save as Draft” instead of “Complete Item” 4a. Merchant Presses “Save as Draft” 4b. System Validates input for correct inputs and skips missing inputs 4c. System saves item as a draft 4d. merchants draft list is updated by the system

Alternative Flow 2: Save as Draft(Item already in drafts)	**Trigger:** User Presses “Save as Draft” instead of “Complete Item” 4a. Merchant Presses “Save as Draft” 4b. System Validates input for correct inputs and skips missing inputs 4c. Dialog Box pops up “Name already exists in drafts would you like to update it” 4d. User Presses Yes name stays the same info updated 4e. user presses No User picks new name 4f. System saves item as a draft 4g. merchants draft list is updated by the system
Exception Flow 1: Invalid Price	**Trigger:** Price not > $0 or is not in numerical format 3a. Error Message Is Displayed 3b. price input box highlighted 3c. Merchant corrects price 3d. Flow returns to step 4 of Main Flow
Exception Flow 2: Invalid Item Name	**Trigger:** Item name already appears in merchant’s menu 3a. Error Message Is Displayed 3b. Name Input Box highlighted 3c. merchant corrects name 3d. flow returns to step 4 of Main flow

Exception Flow 3: System Error prevents adding item	**Trigger:** System fails at some point in process not fully completing 4a. Error Message displays “System Error Item Not Created” 4b. Any previous parts of the process are undone (All or nothing function)
Exception Flow 4: Invalid Calorie Input	**Trigger:** Item calorie < 0 3a. Error Message Is Displayed 3b. calorie input box highlighted 3c. Merchant corrects calories 3d. Flow returns to step 4 of Main Flow
Business Rules	· Item name must be unique within a merchants menu · Price must be positive · Calories must be positive · Draft Items won’t appear on live menu · All fields must be validated before being saved as a menu item
Frequency of Use	High – Used whenever there is a change on Menu
Priority	High- Core Functions of Store still functional but function is needed for improved merchant experience

Use Case ID	UC-M04
Use Case Name	Confirm/Delete Orders

Actor	Registered Merchant
Description	Allow Merchant's to control what orders they accept
Pre-conditions	Merchant Emailed Verified Store Status is Open Merchants menu has at least one item currently available
Post-conditions(success)	Order is moved to out of new orders page into current orders queue
Post-conditions(Failure)	Order does not move from new orders page Error Message displayed
Main Flow(Basic Flow)	1. Order appears on new orders page 2. Merchnant presses the "Confirm Order" button 3. System moves order from new orders page to current orders queue 4. System sends out order confirmed notification to customer
Alternative Flow 1: User Presses Cancel Order	**Trigger:** User presses Cancel instead of confirm 2a. Merchant presses cancel order 2b. Order is removed from live operations page 2c. Customer is notified order is canceled.
Business Rules	· Store must be open for orders to come in · Any changes in order status should be sent to the customer
Frequency of Use	High-Used Whenever the Merchant is actively selling

Priority	Critical-Needed for core functionality of app

Use Case ID	UC-M05
Use Case Name	View Reports
Actor	Registered Merchant
Description	Allow users to view reports on orders they have completed
Pre-conditions	Merchant is logged in Merchant has completed at least one order
Post-conditions(success)	Report is generated
Post-conditions(Failure)	Error Message displays
Main Flow(Basic Flow)	1. User selects daily, monthly or yearly reports reports from selection 2. User Selects either income or item sales reports 3. Report is generated 4. The report is displayed on screen
Exception Flow 1: System Failure	**Trigger:** System fails while generating report 3a. report is not generated 3b. Display error message on screen
Business Rules	To access reports at least one
Frequency of Use	Medium- Useful for merchants business
Priority	Medium- Not needed for core functionality can be put off for other more important functions

Use Case ID	UC-M06
Use Case Name	Edit Restaurant Information
Actor	Registered Merchant
Description	Allow merchant to edit the information of their restaurant
Pre-conditions	Merchant is logged in Merchant is on the merchant dashboard
Post-conditions(success)	Merchant Information is updated
Post-conditions(Failure)	Error Message displays
Main Flow(Basic Flow)	1. User clicks on the 'edit' button to enter edit mode. 2. User edit the information of any of the following fields: 2a. Restaurant name 2b. Restaurant address 3. User clicks on the 'save' button 4. User information is updated with the screen rendered
Exception Flow 1: System Failure	**Trigger:** System fails while save the changes 4a. User information is not updated 4b. Display error message on screen
Exception Flow 2: Incorrect Input	**Trigger: User leaves the fields empty.** 4a. User information is not updated 4b. Display error message on screen
Business Rules	N/A

Frequency of Use	Medium- Useful for merchants business
Priority	Medium- Not needed for core functionality can be put off for other more important functions

IV. Use Case to Screen Mapping

Use Case	Web Page	Description
UC-G01	login.html	The Email/Password fields and the "Sign In" button handle credential entry and submission (steps 3-4).
UC-G02	register.html	The fields for required information and the 'Register' button handle entering details, selecting a role, and submitting the new account (steps 3-6)
UC-C01	browse.html, restaurant-details.html	The search bar handles keyword entry (step 3), the Search button executes the query (step 4), and the displayed listings allow selection (step 6).
UC-C02	cart.html	The Cart page displays items. The "+" and "−" buttons adjust quantities (step 3), triggering the price recalculation (step 4).
UC-C03	checkout.html, order-confirmation.html	The 'Proceed to Checkout' button initiates the checkout page. The address input fields and the 'Place Order' button handle delivery details and order finalization (steps 1, 3, 6).
UC-C04	customer-dashboard.html, customer-order-status.html	The Order History section displays the list of previous orders (step 3), and clicking an order link displays the detailed information (step 5).
UC-C05	profile.html	The 'Edit' button initiates the flow (step 2). The editable fields (Street, City, State, etc) and the 'Save' button handle the address update (steps 3-5).
UC-M01	my-menu.html	The add button function handles step 1 of the main flow, form validation will handle Exception flow 1 2 and 4.

UC-M02	my-menu.html	The delete button function handles step 1 of the main flow.
UC-M03	my-menu.html	The edit button function handles step 1 of the main flow, form validation will handle Exception flow 1 2 and 4.
UC-M04	live-operations.html	The Pending orders section handles step 1 of basic flow with the confirm order button handling step 2, Alternative Flow 1 is handled by the “Cancel” button
UC-M05	merchant-reports.html	The tabs allow user to choose which type of report they want and Handles Step 1 and 2 of the main flow
UC-M06	live-operations.html	The ‘edit’ button handles step 1 of entering edit mode and the ‘save’ button handles step 3 of saving the changes.